Khushaan Virk

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Technical Skills

Embedded & Control Systems: 8051 (N76E003, EFM8LB1, CV-8052), ARM (STM32), FPGA, ADC/DAC, I2C, SPI, UART, PWM
Programming: C, Embedded C, Assembly (8051), Python, SQL, SystemVerilog
Instrumentation & Hardware: Oscilloscopes, multimeters, soldering, signal conditioning, op-amp design, PCB layout
Tools & Software: Git, KiCad, EasyEDA, MATLAB, serial communication (PuTTY), CrossIDE

Education

University of British Columbia — Vancouver, BC
Bachelor of Applied Science (BASc) in Electrical Engineering — *Expected May 2028*
- **Relevant Coursework:** Circuit Analysis, Digital Logic Design, Signals & Systems, Embedded Systems Design

Experience

Software Engineer — Jan. 2026 – Present
Advizr — *Vancouver, BC*
- Designed and shipped end-to-end Python automation systems for clients, including web scraping, REST API integrations, SQL storage, and automated email pipelines, working from open-ended goals rather than fixed specs
- Picked up new libraries and APIs on a per-project basis by reading documentation directly, applying the same approach used to learn microcontroller peripherals from datasheets
- Diagnosed production issues in deployed client systems by reproducing failures locally, isolating the root cause, and verifying fixes before redeployment
- Maintained client systems across ongoing change requests using Git for version control, balancing new feature work against keeping existing deployments stable

Sports Instructor / Summer Camp Leader — Sep. 2023 – Sep. 2025
City of Surrey — *Surrey, BC*
- Led structured programs for groups of 15–30, coordinating daily operations across staff and volunteers while maintaining safety standards
- Adapted instruction in real time to participants of varying skill levels, communicating clear expectations to children, parents, and co-instructors while resolving conflicts and incidents on the spot
- Managed concurrent daily responsibilities including equipment setup, attendance tracking, lesson delivery, and end-of-day reporting under fixed program schedules

Projects

Field-Following Autonomous Robot | *EFM8, STM32, C, MOSFETs, I2C, IR* — Apr. 2026
- Built a dual-microcontroller robot (8051 + ARM) with MOSFET H-bridge motor drivers and opto-isolated control lines for electromagnetic noise reduction during inductive sensor readings
- Implemented magnetic field detection using pick-up inductors to autonomously follow a guide wire, with a third sensor for intersection detection and pre-configured path navigation
- Designed an IR remote controller on a separate MCU transmitting 38kHz modulated commands for manual control and path selection
- Integrated VL53L0X Time-of-Flight laser ranging modules over I2C for real-time collision detection and avoidance in autonomous mode

Reflow Oven Controller | *CV-8052, Assembly, Op-Amp, ADC, SSR, Python* — Feb. 2026
- Designed a closed-loop temperature control system using a K-type thermocouple with cold-junction compensation (LM335) and OP07 op-amp signal conditioning, achieving ±3°C accuracy over 25–240°C
- Wrote interrupt-driven assembly firmware implementing a 5-state FSM with PWM-based SSR control for precise thermal profile tracking through preheat, soak, reflow, and cooling phases
- Built a Python serial data acquisition dashboard with real-time strip-chart plotting and automated Excel report generation over UART at 115200 baud
- Integrated LCD/7-segment displays, speaker feedback, and safety logic with automatic abort on sensor failure

AC Phasor Voltmeter | *EFM8LB1, C, ADC, Signal Processing* — Nov. 2025
- Designed a microcontroller-based AC instrument displaying RMS voltage and phase difference between two 60Hz sinusoidal inputs using high-speed ADC sampling and digital signal processing in C

Capacitance Meters (Dual Platform) | *EFM8LB1, STM32, 555 Timer, C* — Oct.–Nov. 2025
- Built two capacitance meters on different MCU families (8051 and ARM) using 555 timer astable oscillators for frequency-based capacitance measurement (1nF–1μF); designed the 555 equivalent from discrete comparators, SR flip-flop, and transistor discharge circuit